Bandwidth and Transmission rate

Bandwidth

- **Bandwidth** refers to the maximum amount of data that can be transmitted over a communication channel in a given amount of time.
- It is typically measured in bits per second (bps) or hertz (Hz) and represents the capacity of the channel to carry data.
- A wider bandwidth allows for the transmission of more data simultaneously, resulting in higher data rates and better overall performance.
- Bandwidth is not the same as transmission rate; the transmission rate represents the actual speed at which data is transmitted, while bandwidth refers to the channel's capacity to carry data.

Bandwidth (continue…/)

- **Factors affecting bandwidth**

1. **Physical Properties of the Communication Medium**:
   - The physical properties of the medium through which the communication signals are transmitted play a crucial role in determining the bandwidth.
   - For example, in wired systems, the characteristics of the cables, such as their thickness, material, and resistance, affect the data-carrying capacity.
   - In wireless systems, factors like frequency range, available spectrum, and environmental conditions (e.g., interference, obstacles) impact the achievable bandwidth.
2. **Technology Being Used**:
   - Different communication technologies have varying bandwidth capabilities.
   - For instance, older dial-up modems had limited bandwidth capacities, typically around 56 Kbps.
   - On the other hand, modern technologies like fiber optics, DSL (Digital Subscriber Line), cable, and wireless technologies such as 4G/5G provide much higher bandwidths, ranging from Mbps to Gbps.
   - The specific technology used determines the maximum data rates that can be achieved.

Bandwidth (continue…/)

- **Factors affecting bandwidth (continue../)**

3. Signal Processing Techniques:

- Signal processing techniques can enhance the effective bandwidth by optimizing data transmission.
- Techniques like modulation, coding, and compression are used to encode and compress data, allowing more information to be transmitted within the available bandwidth.

4. Network Congestion:

- Bandwidth can also be influenced by network congestion, especially in shared networks.
- When multiple users or devices are accessing the same network simultaneously and generating high traffic, the available bandwidth is divided among them.
- This can lead to slower data transfer rates and reduced overall bandwidth for each user

Bandwidth (continue…/)

- **Factors affecting bandwidth (continue../)**

5. Quality of Service (QoS) Requirements:

- In some cases, specific applications or services may require a certain level of bandwidth to function optimally.
- For example, real-time applications like video conferencing or online gaming demand higher bandwidth to ensure smooth and uninterrupted data transmission.
- In such cases, network administrators may allocate dedicated bandwidth or prioritize certain traffic to meet the QoS requirements.

Transmission rate

- **Transmission rate** refers to the speed at which data can be transmitted over a communication channel or medium.
- It is usually measured in bits per second (bps) and represents the number of bits that can be transmitted in one second.
- Transmission rates can vary widely depending on the technology used communication medium being used.
- Higher transmission rates allow for faster data transfer and better overall performance in communication systems.

Transmission rate (continue…/)

Factors affecting transmission rate

1. **Quality of the communication channel:**
   - The quality of the communication channel refers to the overall condition and reliability of the medium through which data is transmitted.
   - Factors such as signal attenuation, noise, and distortion can affect the quality of the channel.
   - Signal attenuation occurs when the strength of the signal diminishes as it travels through the channel, resulting in a decrease in transmission rate.
   - Noise refers to unwanted signals or disturbances that can interfere with the original signal, leading to errors and reducing the effective transmission rate.
   - Distortion occurs when the shape or characteristics of the transmitted signal are altered during transmission, which can degrade the data and impact the transmission rate.
2. **Interference:**
   - Interference refers to the presence of unwanted signals or electromagnetic radiation that can disrupt or degrade the original signal.
   - Interference can be caused by various sources, such as other electronic devices, power lines, or even natural phenomena like lightning.
   - When interference occurs, the quality of the received signal can be compromised, leading to errors and a decrease in the effective transmission rate.
   - To mitigate interference, techniques such as shielding, filtering, and error correction codes are employed to improve the transmission rate and maintain data integrity.

Transmission rate (continue…/)

Factors affecting transmission rate

3. Distance between the sender and receiver:

- The distance between the sender and receiver can affect the transmission rate due to signal degradation over distance.
- As the distance increases, the strength of the signal diminishes, leading to signal attenuation and a decrease in the effective transmission rate.
- To overcome the limitations posed by distance, amplification and signal regeneration techniques may be used to boost the signal strength and maintain an acceptable transmission rate.
- In long-distance communication, the use of repeaters or signal amplifiers at regular intervals helps to extend the reach and maintain the desired transmission rate.

Relationship between Transmission rate and Bandwidth.

- The transmission rate and bandwidth are related but not interchangeable terms.
- The transmission rate is a measure of the actual speed at which data is transmitted, while bandwidth refers to the channel's capacity.
- The transmission rate cannot exceed the bandwidth of the channel, as it is limited by the channel's capacity.
- However, the transmission rate can be lower than the bandwidth if the channel is not fully utilized or if there are other limiting factors.
- In practical terms, achieving higher transmission rates requires a communication medium with a wider bandwidth or the use of more advanced modulation techniques to increase the data-carrying capacity within the available bandwidth.

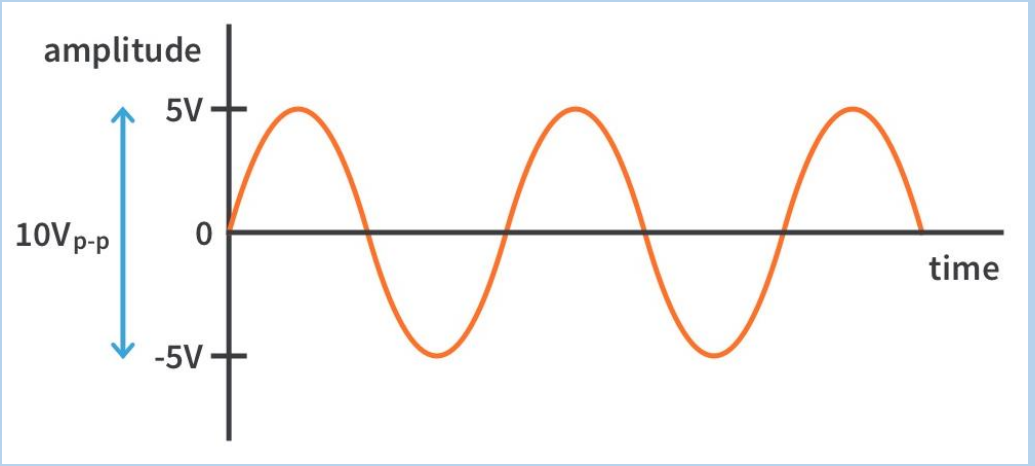
Signals, types of signals and standards

What are signals?

- **Signals** are electrical, electromagnetic, or optical representations of data or information.
- They are used to transmit, convey, or store information in various systems, including communication, electronics, and computing.
- Signals can take different forms and can be classified based on their characteristics and properties.

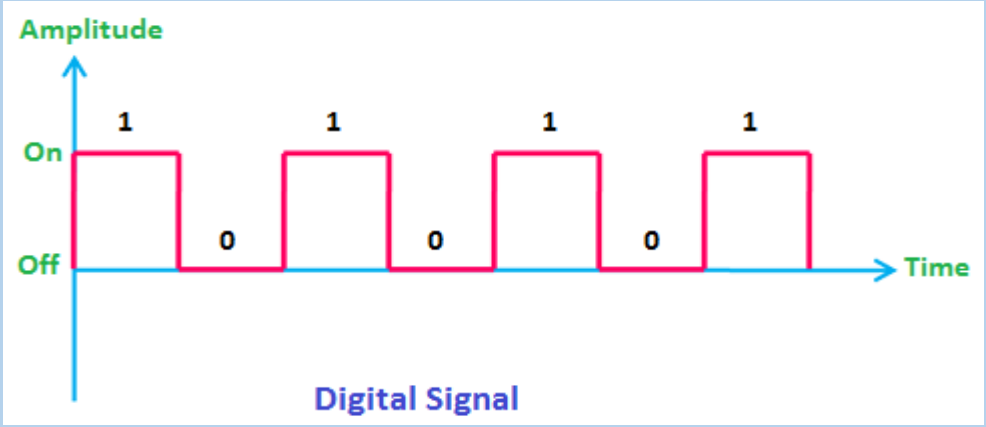
Types of signals

1. Analog Signals



- Analog signals are continuous signals that represent information using varying physical quantities such as voltage, current, or frequency.
- They can take on any value within a specific range and are used to convey real-world phenomena, such as sound, temperature, or pressure.
- Analog signals are susceptible to noise and distortion during transmission, which can degrade the quality of the signal.

Types of signals …



2. Digital Signals

- Digital signals are discrete signals that represent information using binary digits (bits), which can take on only two possible values: **0** and **1**.
- They are more resistant to noise and distortion compared to analog signals, as they can be accurately regenerated.
- Digital signals are widely used in modern communication systems, computer networks, and digital devices.

Signal Standards

- Signal standards, also known as communication standards or protocols, define the rules, specifications, and formats that govern the transmission, reception, and interpretation of signals in a specific system or application.
- These standards ensure interoperability, compatibility, and efficient communication between devices and systems.
- Some of these standards include:

Signal Standards…

1. **Communication Standards**
   - Communication standards define the protocols, formats, and procedures required for devices to communicate with each other.
   - Standards ensure interoperability and compatibility between different devices and systems.
   - Examples of communication standards include *Ethernet (IEEE 802.3)*, *Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15.1), and GSM (Global System for Mobile Communications).*
2. **Data Transmission Standards**
   - Data transmission standards specify the rules and specifications for transmitting data over various communication channels.
   - They define the physical and electrical characteristics of the transmission medium, modulation techniques, error detection and correction methods, and data framing.
   - Examples of data transmission standards include *RS-232, USB (Universal Serial Bus), HDMl (High-Definition Multimedia Interface), and Ethernet (IEEE 802.3).*

Signal Standards ….

3. **Internet and Networking Standards:**
   - Internet and networking standards govern the protocols and technologies used for communication and data exchange over computer networks.
   - These standards enable the seamless transfer of data between different networks and devices.
   - Examples of internet and networking standards include
     - TCP/IP (Transmission Control Protocol/Internet Protocol)
     - HTTP (Hypertext Transfer Protocol)
     - DNS (Domain Name System)
     - Wi-Fi (IEEE 802.11).

Transmission media

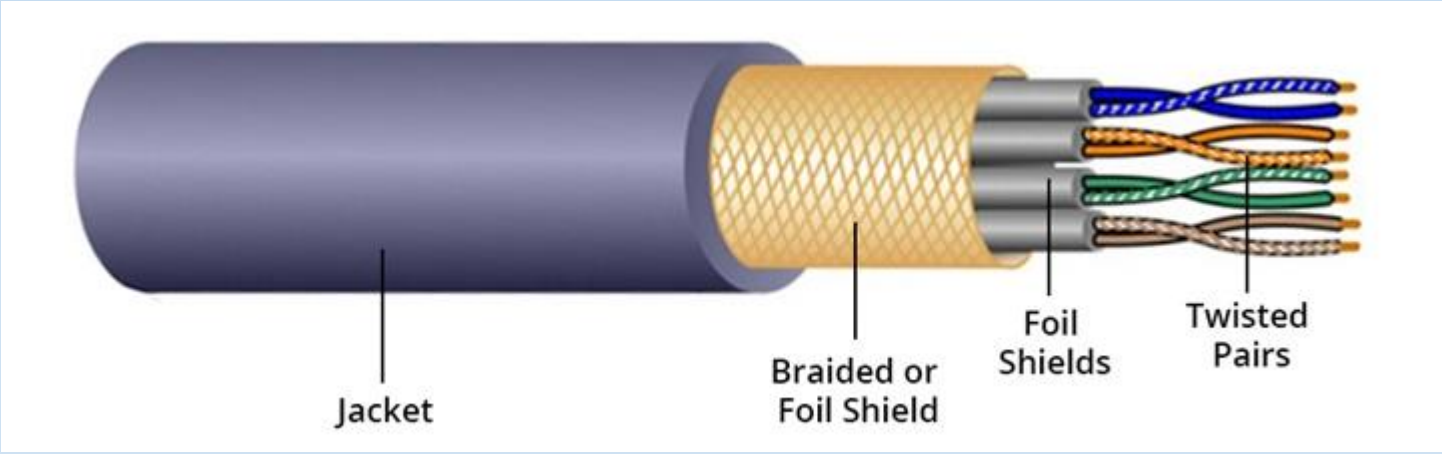
What is transmission media

- **Transmission media**, also known as communication channels or communication media, refer to the pathways or mediums through which signals are transmitted from one location to another.
- Transmission media can be categorized into two main types based on their physical nature:

1. Guided media
2. Unguided media

Guided transmission media

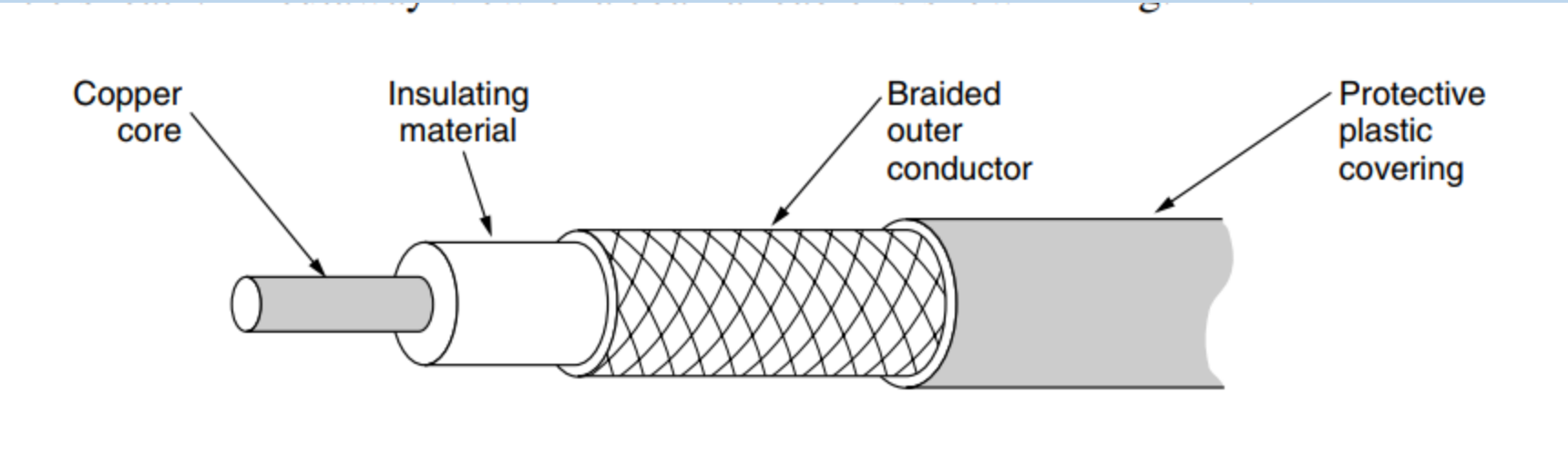
- Guided media, also known as **bounded** or **wired** media, refer to transmission media that provide a physical pathway for signals to travel through.
- Examples of guided media include:
    1. **Twisted Pair Cable:** Consists of pairs of copper wires twisted together and is commonly used for telephone lines and Ethernet connections.



Guided transmission media…

2. Coaxial Cable

- Consists of a central conductor, an insulating layer, a metal shield, and an outer insulating layer.
- It is used for cable television (CATV) and high-speed data transmission.

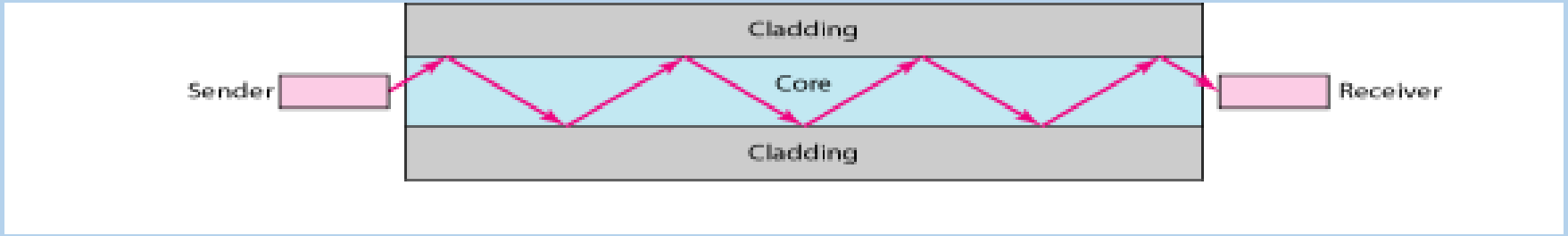


Guided transmission media…

3. Fiber Optic Cable

- A fiber optic cable is made up of glass or plastic and transmits signals in the form of light.
- Light travels in a straight line as long a it is moving through a single uniform substance.
- Optical fiber use reflection to guide light through a channel.
- Fiber optics are used for long-haul transmission in network backbones, high speed LANs.

- An Optical Transmission System has three key components:
  1. **The light source –** it generates light signal that will be transmitted through the fiber optic cable.
  2. **Transmission medium –** Ultra-thin fiber of glass or plastic, it is made up of a core (where light travels) and cladding (which reflects the light back)
  3. **The detector –** This is responsible for converting the light signal back into electrical signal that can be processed by electronic devices. The most common detector is **photodiode.**



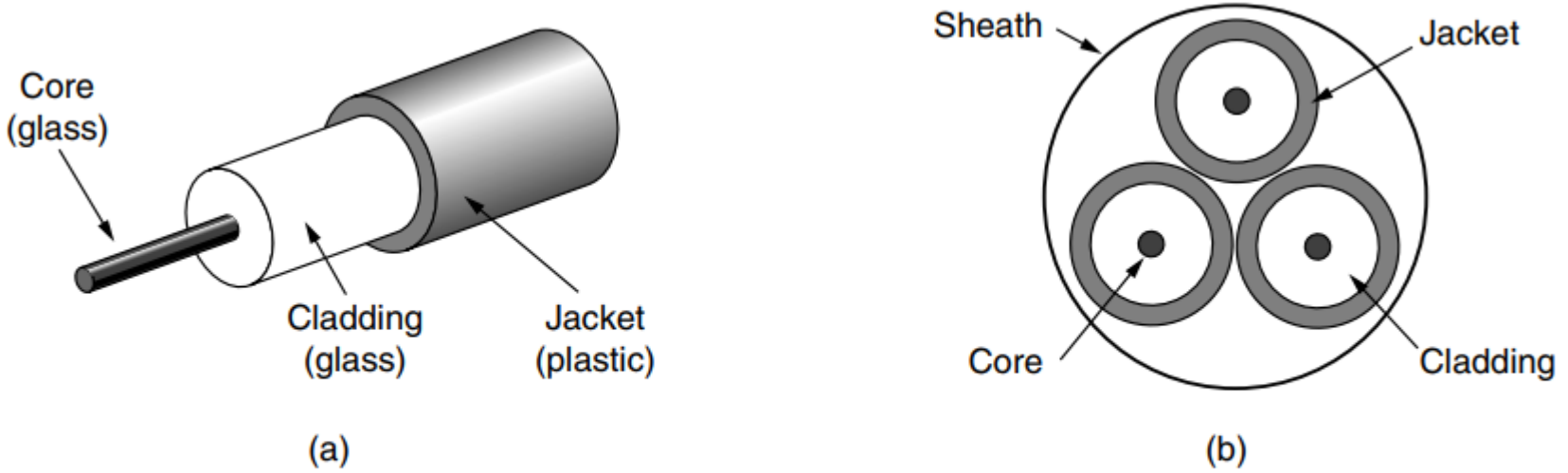


Figure 2-8. (a) Side view of a single fiber. (b) End view of a sheath with three fibers.

Unguided media

- Unguided media rely on wireless transmission through the air or space.
- They offer greater flexibility but facing challenges like signal attenuation and susceptibility to atmospheric conditions.
- Unguided media are also known as **wireless** or **unbounded** media.

- Examples include:
  - 1. **Radio Waves:**
    - Used in wireless communication systems.
    - Includes AM and FM radio, television broadcasting, and Wi-Fi.

2. **Microwaves:**
   - Electromagnetic waves with higher frequencies than radio waves.
   - Used for point-to-point communication, such as in satellite communication and some wireless backhaul links.

3. **Infrared Waves:**
   - Uses infrared light for short-range communication.
   - Common in devices like remote controls and some short-range data transfer technologies.

4. **Satellite Communication:**
   - Involves communication through geostationary or non-geostationary satellites.
   - Used for long-distance and global communication.